THE KOSZUL DERIVED LIMIT FUNCTOR IS THE DERIVED COMPLETION ADJOINT (SECOND PROOF)

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Allowed standard prerequisites (editorial).

- The derived completion adjoint is given by RHom out of the extended alternating Cech complex on f_1, ..., f_r. invoked at proof lines 26406-26413. Provides the explicit model of K^wedge.
- The extended alternating Cech complex is the colimit of the Koszul chain complexes K^n = K(A, f_1^n, ..., f_r^n), whose duals are the Koszul cochain complexes K_n. invoked at proof lines 26414-26428. Rewrites the colimit as a hocolim of perfect complexes with controlled duals.
- For a system of perfect complexes K_n with colimit K, RHom(hocolim K^n, K) = Rlim (K tensor K_n). invoked at proof lines 26432-26433. Concludes the identification of the two functors.

Source reference. Original section 0BKC: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26305–26305. Referenced at statement line 26389 for the setting; the situation is used inside the counted span.

Source reference. Original lemma 091V: sources/stacks-more-algebra/pinned-original-source.tex, source lines 26154–26154. Cited at proof lines 26406-26407 for the explicit model of the adjoint; the conclusion is used inside the counted span.

Source reference. Original lemma 0913: sources/stacks-more-algebra/pinned-original-source.tex, source lines 7100–7100. Cited at proof lines 26414-26416 to express the Cech complex as a colimit of Koszul chain complexes; the conclusion is used inside the counted span.

Source reference. Original lemma 07VI: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19176–19176. Cited at proof lines 26424-26426 for the duals of the Koszul complexes; the conclusion is used inside the counted span.

Source reference. Original lemma 0BKB: sources/stacks-more-algebra/pinned-original-source.tex, source lines 19236–19236. Cited at proof lines 26432-26433 for the final identification of the functors; the conclusion is used inside the counted span.

Lemma 0.2. *In Situation 0BKC. The functor which sends $K \in D(A)$ to the derived limit $K' = R\lim(K \otimes^{\mathbf{L}}_A K^\bullet_n)$ is the left adjoint to the inclusion functor $D_{comp}(A) \to D(A)$ constructed in Lemma 091V.*

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Second proof. Denote $K \mapsto K^\wedge$ the adjoint constructed in Lemma 091V. By that lemma we have

$$K^\wedge = R\mathrm{Hom}\left((A \to \prod\nolimits_{i_0} A_{f_{i_0}} \to \prod\nolimits_{i_0<i_1} A_{f_{i_0}f_{i_1}} \to \ldots \to A_{f_1\ldots f_r}), K\right)$$

In Lemma 0913 we have seen that the extended alternating Čech complex

$$A \to \prod\nolimits_{i_0} A_{f_{i_0}} \to \prod\nolimits_{i_0<i_1} A_{f_{i_0}f_{i_1}} \to \ldots \to A_{f_1\ldots f_r}$$

is a colimit of the Koszul complexes $K^n = K(A, f_1^n, \ldots, f_r^n)$ sitting in degrees $0, \ldots, r$. Note that K^n is a finite chain complex of finite free A-modules with dual (as in Lemma 07VI) $R\mathrm{Hom}_A(K^n, A) = K_n$ where K_n is the Koszul cochain complex sitting in degrees $-r, \ldots, 0$ (as usual). Thus it suffices to show that

$$R\mathrm{Hom}_A(\mathrm{hocolim} K^n, K) = R\lim(K \otimes^{\mathbf{L}}_A K_n)$$

This follows from Lemma 0BKB. □